

59. (d) The National Research Centre for Agroforestry (NRCAF), as a unit of ICAR was established on 8 May 1988. The National Research Centre for Agroforestry (NRCAF), which was renamed the Central Agroforestry Research Institute ) on 1 December 2014, is located in Jhansi. This institute was established under the Indian Council of Agricultural Research (ICAR) to conduct research and development in agroforestry.
60. (b) Lotia is a type of regional music from Rajasthan, typically sung by farmers while they work in the fields. It is especially popular during the Lotia festival, which takes place in the month of Chaitra according to the Hindi calendar. This festival and the singing occur in early spring in Rajasthan.
61. (d) World Earth Day, celebrated on April 22nd each year, is a global event dedicated to environmental protection. First observed in 1970, it now involves activities and events coordinated worldwide by EarthDay.org, including 1 billion people in more than 193 countries. The official theme for 2024 is "Planet vs. Plastics. The concept of Earth Day was first introduced at a UNESCO Conference in San Francisco. Additionally, the well-known Paris Agreement, which focuses on climate change, was signed by the United States, China, and around 120 other countries on Earth Day in 2016.
62. (d) Article 360 of the Indian Constitution provides the provision for declaring a financial emergency. This article allows the President of India to declare a financial emergency if he/she believes that the financial stability or credit of India or any part of its territory is threatened. A declaration of financial emergency must be confirmed by both the Lok Sabha and the Rajya Sabha within two months from the date it is issued.
63. (d) The term of the Prime Minister is not fixed and depends on the pleasure of the President, which means that the Prime Minister holds office as long as he has the confidence of the Lok Sabha. The constitution mandates that the President should appoint as Prime Minister the leader of the majority party in the Lok Sabha or a person who can win the confidence of the Lok Sabha by gaining the support of other political parties. The Constitution allows the Prime Minister to be a member (either elected or

nominated) of either house of Parliament, which includes nominated members of the Rajya Sabha. Therefore, a nominated member of the Rajya Sabha can become the Prime Minister.

Hence, statement 1, 2 and 3 are incorrect.

64. (c) A bolometer is a device used to measure the power of incoming electromagnetic radiation, especially in the infrared spectrum.

It works by utilizing the principle that the electrical resistance of certain materials changes when they absorb radiation.

Bolometers are often used in astronomy to detect and measure weak radiation from space objects.

The term "bolometer" comes from the Greek words "bolos," meaning beam or ray, and "meter," meaning to measure.

65. (d) AIRAWAT, an AI supercomputer, ranked 75th in the Top 500 Global Supercomputing List, affirming India's strong position in AI supercomputing globally. Therefore, statement I is correct.

AIRAWAT is part of India's National Program on AI and has a peak compute capacity of 410 AI Petaflops Mixed Precision, verifying statement II as correct. The Ministry of Electronics and Information Technology (MeitY) plans to enhance AIRAWAT's capabilities to 1,000 AI Petaflops Mixed Precision to support India's growing AI needs, making statement III correct.

In summary, all three statements regarding AIRAWAT's ranking, capabilities, and future plans are correct.

66. (a) The Barak 8 is a long-range surface-to-air missile created by Israel Aerospace Industries and India's DRDO.

It was first successfully tested in Israel on November 10, 2014.

The Indian Navy also conducted successful tests of the missile from the INS Kolkata on December 29 and 30, 2015.

Israel successfully tested an extended-range version of the Barak 8 missile on March 22, 2021.

67. (d) The "Kavach" system introduced by Indian Railways is an Automatic Train Protection System.

The KAVACH is an indigenously developed Automatic Train Protection (ATP) system by the Research Design and Standards Organisation (RDSO) in collaboration with the Indian industry.

The Union Railway Minister Ashwini Vaishnaw inspected the trial of the Kavach working system between Gullaguda-Chitgidda Railway stations on Lingampalli-Vikarabad section in the Secunderabad

Division of South Central Railway in March 2023.

68. (b) The capital of Bulgaria is Sofia.

Located in the western part of Bulgaria, at the foot of the Vitosha mountain, Sofia is one of the oldest cities in Europe with a history that stretches over 7,000 years.

It's known for its unique blend of Eastern and Western cultures, evident in its architecture which includes a mix of Ottoman, Soviet, and modern European influences.

69. (c) The winner of the 2023 International Booker Prize is from Bulgaria.

Georgi Gospodinov, a Bulgarian writer, won the prize for his novel "Time Shelter," marking a significant achievement as this was the first time a Bulgarian novel had won this prestigious award.

70. (c) Vistara Airlines is a joint venture between Tata Sons and Singapore Airlines, based in Gurgaon, India. It was established in 2013 in New Delhi. Vistara operates within India.

Air Pegasus was a regional airline based in Bangalore, India, founded in 2014. It primarily served domestic routes.

Air Asia India, headquartered in Bangalore, Karnataka, was founded by Tony Fernandez. This airline is known for offering an advanced food ordering service to make in-flight meals more affordable compared to typical airport prices.

Turbo Aviation Private Limited, founded in November 2003, started Turbojet Airways, a regional airline based in Hyderabad, on March 4, 2013. This airline serves regional destinations within India.

71. (d) The Ashok Chakra is India's highest peacetime gallantry award, ranking above the Kirti Chakra and Shaurya Chakra.

It is given for acts of bravery and self-sacrifice away from the battlefield.

The first recipient of the Ashok Chakra was Flight Lieutenant Suhas Biswas in 1952.

On the other hand, the Vir Chakra is a wartime gallantry award in India, third in order of precedence after the Param Vir Chakra and the Maha Vir Chakra.

It is awarded for acts of bravery in the presence of the enemy, whether on land, at sea, or in the air. The first recipient of the Vir Chakra was Brigadier Harbaksh Singh in 1948.

72. (c) India bought two Phalcon Airborne Warning and Control Systems (AWACS) from Israel for approximately USD 1 billion.

AWACS are advanced mobile radar systems that provide long-range air defense surveillance.

They are often referred to as an “eye in the sky” because of their extensive surveillance capabilities.

73. (c) Nag Missile- It is primarily an anti-tank guided missile, which means it is designed for surface-to-surface engagements, specifically targeting armored vehicles like tanks. It is not used for air-to-surface purposes.

Barak missile - ship-to-air, ship-to-surface: This is correctly matched as the Barak missile family includes variants capable of engaging targets in both air and surface scenarios from ships.

Nirbhay Missile - Naval, Land, Air: This is also correct as the Nirbhay is a long-range, all-weather, subsonic cruise missile that can be launched from multiple platforms including naval ships, land vehicles, and aircraft.

74. (b) Operation Nistar was a humanitarian and disaster relief operation where the Indian Naval Ship (INS) Sunayana evacuated 38 Indian nationals from cyclone-hit Socotra Island in Yemen.

Operation Pawan refers to the Indian Peace Keeping Force's efforts in 1987 to disarm the LTTE and gain control of the Jaffna Peninsula in Northern Sri Lanka.

Operation Karuna was an aid operation by India to help Myanmar after Cyclone Mocha, where Indian Navy ships delivered over 40 tonnes of relief materials.

Operation Devi Shakti involved the evacuation of people, including Indians, from Taliban-controlled Kabul to Dushanbe.

Therefore, the correct matches for List I to List II based on the detailed descriptions are:

I-A, II-B, III-C, IV-D.

75. (c) The High Altitude Warfare School (HAWS) of the Indian Army, located in Gulmarg, Jammu and Kashmir, is a premier training institute specializing in mountain warfare and high altitude operations.

Established initially at Sonamarg in 1948, it was later moved to Gulmarg in 1962.

On April 8, 1962, the school was classified as a Category A Training Establishment and was renamed the High Altitude Warfare School (HAWS).

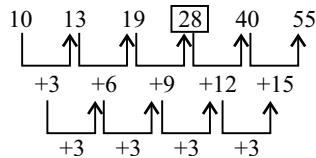
Soldiers who are posted to the Siachen Glacier and other high-altitude areas receive their training at the High Altitude Warfare School (HAWS).

76. (d) Saxophone, Clarinet, and Trumpet are all wind instruments that produce sound

from the vibration of air. Drum, however, is a percussion instrument that produces sound from being struck.

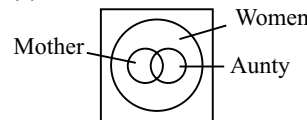
77. (b) A Station is a place where trains regularly stop to load or unload passengers and goods. Similarly, A Garage is a building or indoor area commonly used to store a car safely when not in use.

78. (a) The pattern followed is:

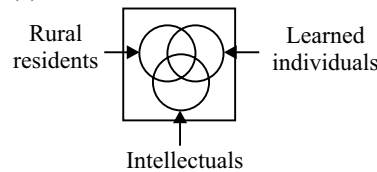


79. (b) Basketball is played on a court. Similarly, Boxing is conducted in a ring.

80. (a)



81. (b)



82. (b) Logic: format ' $x - x^2$ '

$$23 - 23^2 = 529$$

$$29 - 29^2 = 841 \neq 831$$

$$27 - 27^2 = 729$$

$$31 - 31^2 = 961$$

83. (c) Second row  $\rightarrow (9 + 4)^2 = 13^2 = 169$

$$\text{Third row} \rightarrow (11 + 10)^2 = 21^2 = 441$$

$$\text{Therefore, First row} \rightarrow (7 + 8)^2 = 15^2 = 225$$

84. (d) The logic is:

$$31 \times 3 - 3 = 90$$

$$37 \times 5 - 5 = 180$$

Similarly,

$$? \times 7 - 7 = 287$$

$$\Rightarrow ? \times 7 = 287 + 7$$

$$\Rightarrow ? = 42$$

85. (a)  $15 + 2 = 17$ ,

$$17 + 4 = 21,$$

$$21 + 2 = 23,$$

$$23 + 4 = 27,$$

$$27 + 2 = 29,$$

86. (d) The logic followed here is: In a row:

$$\text{2nd number} = (\text{First Number} \times 3) + 2,$$

$$\text{3rd number} = (\text{First Number} \times 5) + 4$$

$$\text{For Row 1} \rightarrow 7, 23, 119$$

$$(7 \times 3) + 2 = 23, (23 \times 5) + 4 = 119$$

$$\text{For Row 3} \rightarrow 15, 47, 239$$

$$(15 \times 3) + 2 = 47, (47 \times 5) + 4 = 239$$

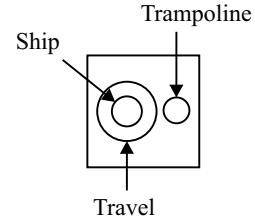
Similarly,

$$\text{For Row 2} \rightarrow 11, 35, ?$$

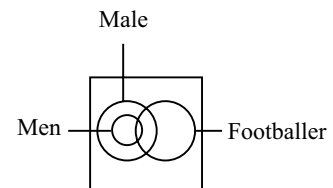
$$\text{2nd number} = (11 \times 3) + 2 = 35$$

$$\text{3rd number} = (35 \times 5) + 4 = 179$$

87. (a)



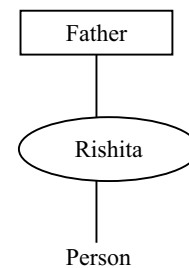
88. (a)



89. (a) 2197, 1331 and 4913 are cubes of 13, 11 and 17, which are prime numbers. Whereas, 3375 is the cube of 15, which is a composite number.

90. (c)

| Symbol in Diagram | Meaning                    |
|-------------------|----------------------------|
| ○                 | Female                     |
| □                 | Male                       |
| ==                | Married couple             |
| —                 | Siblings                   |
|                   | Difference of a generation |



91. (c) Time = 02: 09 = Hours : Minutes

Formula used:

$$\text{Angle} = [(30 \times \text{Hours}) - \{(11 \div 2) \times \text{Minutes}\}]$$

$$\Rightarrow [(30 \times 2) - \{(11 \div 2) \times 9\}]$$

$$\Rightarrow 60 - 49.5 = 10.5$$

92. (a) Time = 02: 09 = Hours : Minutes

$$\text{Angle} = [(30 \times \text{Hours}) - \{(11 \div 2) \times \text{Minutes}\}]$$

$$\Rightarrow [30 \times 9 - \{(11 \div 2) \times 40\}]$$

$$\Rightarrow 270 - 220 = 50$$